ADVANCED EXCEL · LESSON 12 OF 16

Power Query, Power Pivot, Analysis ToolPak & What-If Analysis

Automate cleaning with Power Query (import, transform, unpivot, merge, append, group), model several tables with Power Pivot and DAX, get descriptive statistics and regression from the Analysis ToolPak, plan with Goal Seek, Data Tables, Scenario Manager and Solver, and meet LAMBDA, Python in Excel and macros.

Live class 9 · ✦ Excel for Microsoft 365, 2026 · Advanced · ⏱ 150 min

By the end of this lesson you can

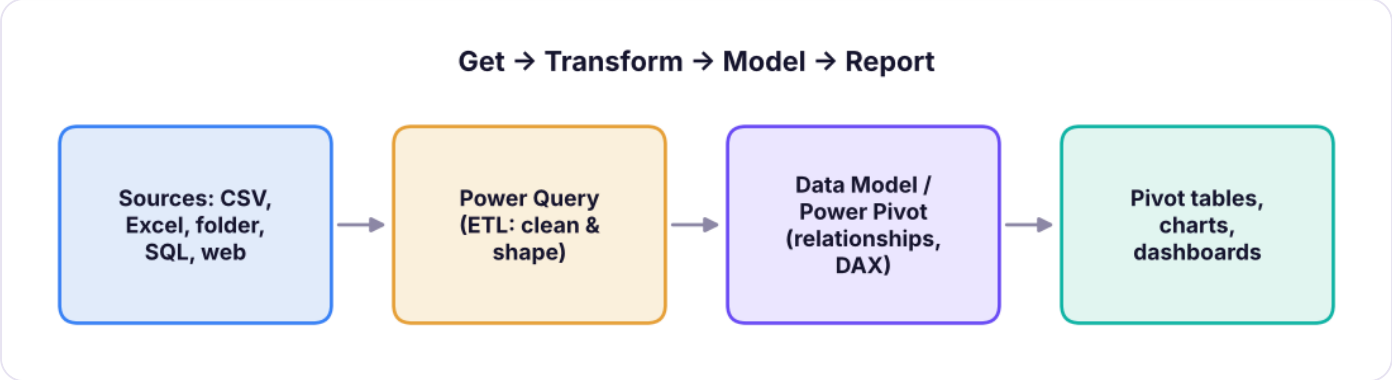
- Build repeatable ETL in Power Query and explain the Applied Steps and M code
- Unpivot wide data, merge (join) and append (union) tables, and group data in Power Query
- Load tables to the Data Model, create relationships and write simple DAX measures in Power Pivot
- Produce descriptive statistics, histograms, correlation and regression with the Analysis ToolPak
- Answer what-if questions with Goal Seek, Data Tables, Scenario Manager and Solver
- Know when to use LAMBDA, Python in Excel, VBA macros and Office Scripts

Datasets used

`superstore_sales.csv` `video_game_sales.csv`

Why "Power" tools?

Formulas and pivots are great until the same cleaning has to be repeated every month, the data has several related tables, or it grows past a million rows. The Power tools solve exactly these problems, and they are the **same engines used in Power BI**, so everything you learn here transfers directly.



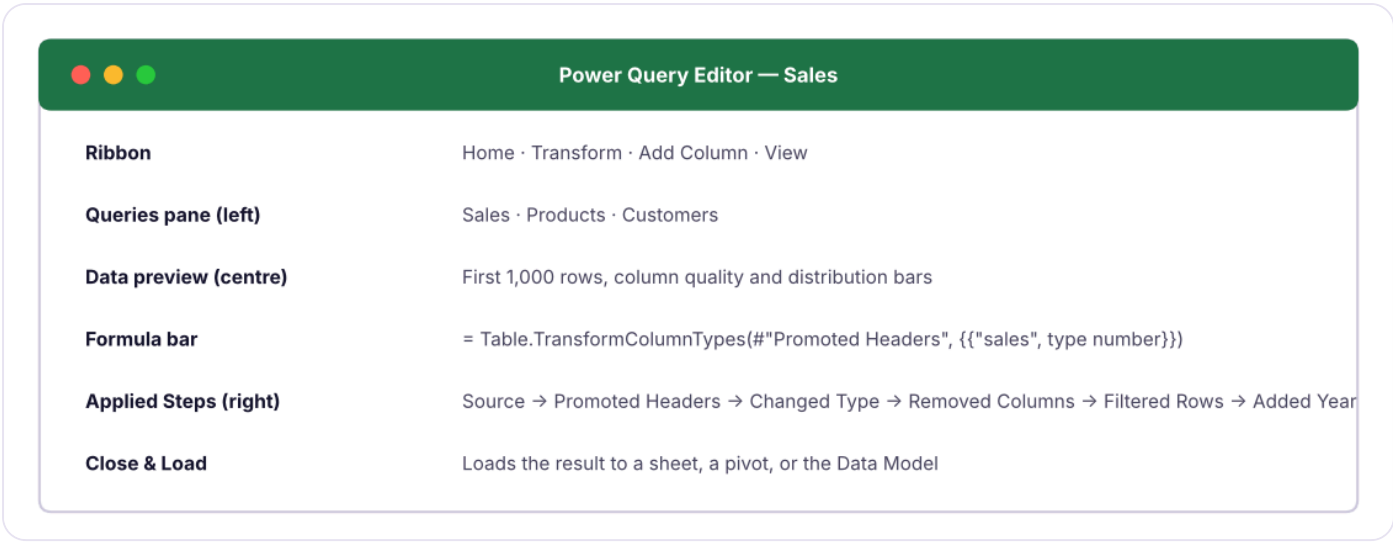
The modern Excel analytics stack. Power Query and the Data Model are shared with Power BI.

Power Query: repeatable cleaning

Power Query (Data → **Get & Transform Data**) imports data and records every cleaning step you click. Next month, put the new file in place and click **Refresh**: all steps run again automatically.

STEP BY STEP **Your first query**

1. Data → **Get Data → From File → From Text/CSV** → choose `superstore_sales.csv` → **Transform Data**. The Power Query Editor opens.
2. Check **data types** in each column header (ABC = text, 123 = whole number, 1.2 = decimal, calendar = date). Fix any wrong ones (right-click header → Change Type).
3. Remove columns you do not need (select → right-click → Remove, or **Choose Columns**).
4. Filter rows with the header drop-downs (for example remove test orders).
5. Add a column: **Add Column → Date → Year**, or **Custom Column** with a formula such as `[sales] - [profit]`.
6. Rename the query (right panel) to `Sales` → **Home → Close & Load To...** → Table, PivotTable report, or **Only Create Connection** + **Add to Data Model**.



The Power Query Editor. Every action you take is recorded in Applied Steps on the right.

Transformations you will use most

Transformation	Where	Example
Remove / keep rows	Home → Remove Rows / Keep Rows	Remove top 3 title rows, remove blank rows, remove duplicates
Use first row as headers	Home → Use First Row as Headers	After importing a report
Split column	Home → Split Column → By Delimiter	`IN-2025-00042` → country, year, number
Replace values	Transform → Replace Values	`Bangalore` → `Bengaluru`
Trim / Clean / Format case	Transform → Format	Same as TRIM, CLEAN, PROPER
Fill down	Transform → Fill → Down	Fill blank category cells (Lesson 2's Go To Special trick, but repeatable)
Conditional column	Add Column → Conditional Column	`if [discount] > 0.2 then "High" else "Normal"`
Column from examples	Add Column → Column From Examples	Like Flash Fill, but repeatable
Group By	Home → Group By	Sales by region (like a pivot)
Unpivot	Transform → Unpivot Columns	Turn month columns into rows
Merge Queries	Home → Merge Queries	A join (lookup) between two tables
Append Queries	Home → Append Queries	Stack tables with the same columns (Jan + Feb + Mar)
From Folder	Get Data → From File → From Folder	Combine every file in a folder automatically

Unpivot: from wide to long

Reports often arrive **wide**: one column per month or per region. Pivot tables, Power BI and Python all prefer **long** data (one row per observation). Unpivot fixes it in one click.

Wide (before): one column per month

	A	B	C	D
1	**Region**	**Jan**	**Feb**	**Mar**
2	North	120	135	150
3	South	98	101	110

STEP BY STEP

Unpivot in Power Query

1. Load the wide table into Power Query.
2. Select the **Region** column (the one that should stay as it is).
3. Transform → **Unpivot Columns** → **Unpivot Other Columns**.
4. Rename `Attribute` to `Month` and `Value` to `Sales`.

Long (after): one row per region and month

	A	B	C
1	**Region**	**Month**	**Sales**
2	North	Jan	120
3	North	Feb	135
4	North	Mar	150
5	South	Jan	98
6	South	Feb	101
7	South	Mar	110

"Unpivot **Other** Columns" is better than selecting the month columns, because new months (Apr, May...) added later are unpivoted automatically. The same operation is `melt()` in pandas and `UNPIVOT` in SQL.

Python

```python
import pandas as pd
wide = pd.DataFrame({"Region": ["North", "South"], "Jan": [120, 98], "Feb": [135, 101],
"Mar": [150, 110]})
wide.melt(id_vars="Region", var_name="Month", value_name="Sales")
```

OUTPUT

	Region	Month	Sales
0	North	Jan	120
1	South	Jan	98
2	North	Feb	135
3	South	Feb	101
4	North	Mar	150
5	South	Mar	110

6 rows × 3 columns

Merge (join) and Append (union)

- **Merge Queries** = a lookup or SQL **JOIN**. Choose the two tables, click the matching key column in each, and choose the join kind: Left Outer (all rows from the first table, matching rows from the second: like XLOOKUP), Inner (only matches), Full Outer, Left Anti (rows with **no** match: great for finding missing products). Then expand the new column to pick which fields to bring in.
- **Append Queries** = SQL **UNION ALL**: stack January, February and March files that share the same columns.

The M language

Every step is written in **M**, Power Query's formula language. View it in Home → **Advanced Editor**:

Power Query M

```
let
    Source = Csv.Document(File.Contents("C:\Data\superstore_sales.csv"), [Delimiter=",",
Encoding=65001]),
    Promoted = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
    Typed = Table.TransformColumnTypes(Promoted, {{"order_date", type date}, {"sales", type
number}, {"profit", type number}}),
    AddedYear = Table.AddColumn(Typed, "Year", each Date.Year([order_date]), Int64.Type),
    HighDiscount = Table.AddColumn(AddedYear, "Discount band", each if [discount] > 0.2
then "High" else "Normal"),
    Grouped = Table.Group(HighDiscount, {"region", "Year"}, {{"Sales", each
List.Sum([sales]), type number}})
in
    Grouped
```

You rarely type M from scratch, but reading it helps you debug, and you can ask an AI assistant to explain or write a step.

> **TIP** **Change the file path once**
>
> Put the file path in a cell, name it `SourcePath`, and in M use `Excel.CurrentWorkbook(){[Name="SourcePath"]}[Content]{0}[Column1]`. Colleagues can then point the query to their own folder without opening the editor.

Power Pivot and the Data Model

The **Data Model** stores tables inside the workbook in a compressed, columnar engine (VertiPaq, the same engine as Power BI). It handles millions of rows and lets one pivot table use **several related tables**, without VLOOKUPs.

> **STEP BY STEP** **Build a small model**
>
> 1. Load `Sales`, `Products` and `Customers` with Power Query → Close & Load To → **Only Create Connection** + **Add this data to the Data Model**.
> 2. Data → **Manage Data Model** (Power Pivot window) → **Diagram View**.
> 3. Drag `Sales[product_id]` onto `Products[product_id]` and `Sales[customer_id]` onto `Customers[customer_id]` to create **one-to-many relationships** (Lesson 2 of the SQL track explains why).
> 4. Insert → PivotTable → **From Data Model**. You can now put `Products[category]` in Rows and `Sales[sales]` in Values: the relationship connects them.

DAX measures in Power Pivot

Measures are formulas that calculate on the fly for whatever is in the pivot (filter context). In Power Pivot → Measures → New Measure (or type in the calculation area):

```DAX
Total Sales := SUM ( Sales[sales] )
Total Profit := SUM ( Sales[profit] )
Profit Margin := DIVIDE ( [Total Profit], [Total Sales] )
Orders := DISTINCTCOUNT ( Sales[order_id] )
Technology Sales := CALCULATE ( [Total Sales], Products[category] = "Technology" )
```

Unlike pivot calculated fields (Lesson 8), `Profit Margin` here is always correct at every level, and `DISTINCTCOUNT` is built in. DAX is covered in depth in the Power BI track (Lesson 4).

Analysis ToolPak: statistics without formulas

The **Analysis ToolPak** is a free add-in: File → Options → Add-ins → Manage: Excel Add-ins → Go → tick **Analysis ToolPak**. It then appears as **Data → Data Analysis**.

Tool	What it produces
Descriptive Statistics	Mean, standard error, median, mode, standard deviation, variance, kurtosis, skewness, range, min, max, sum, count (tick "Summary statistics")
Histogram	Frequency table (and chart) for bins you specify
Correlation	Correlation matrix of several columns
Regression	Linear regression with coefficients, R^2, p-values and residuals
t-Test, z-Test, ANOVA, F-Test	Hypothesis tests (ML track, Lessons 3–5)
Moving Average, Exponential Smoothing	Simple forecasting
Random Number Generation, Sampling	Simulation and sampling

Here is what **Descriptive Statistics** returns for the order-line `sales` column (computed with Python to match the ToolPak's definitions):

Python

```
import numpy as np
from scipy import stats
s = pd.read_csv("superstore_sales.csv")
x = s["sales"]
out = pd.Series({
    "Mean": x.mean(), "Standard Error": x.std() / np.sqrt(len(x)), "Median": x.median(),
"Mode": x.mode().iloc[0],
    "Standard Deviation": x.std(), "Sample Variance": x.var(), "Kurtosis":
stats.kurtosis(x, bias=False), "Skewness": stats.skew(x, bias=False),
    "Range": x.max() - x.min(), "Minimum": x.min(), "Maximum": x.max(), "Sum": x.sum(),
"Count": len(x)}, name="sales")
out.round(2).to_frame()
```

OUTPUT

	sales
Mean	31,854.11
Standard Error	759.76
Median	8,430.00
Mode	420.00
Standard Deviation	60,414.17
Sample Variance	3,649,872,212.91
Kurtosis	16.52
Skewness	3.61
Range	619,248.00
Minimum	42.00
Maximum	619,290.00
Sum	201,413,569.00

13 rows × 1 columns (first 12 shown)

How to read it: the mean (about 31,900) is far above the median (about 8,400), and skewness is strongly positive, so a few very large orders pull the average up. Report the **median** as the typical order line.

The same numbers with Excel functions: `AVERAGE`, `MEDIAN`, `MODE.SNGL`, `STDEV.S`, `VAR.S`, `KURT`, `SKEW`, `MIN`, `MAX`, `SUM`, `COUNT`.

> **NOTE** **ToolPak output is static**
>
> The Analysis ToolPak writes numbers, not formulas. If the data changes, run the tool again. For live results use the equivalent functions.

What-if analysis

Data → What-If Analysis answers "what would happen if..." questions.

Goal Seek: work backwards to a target

"What price gives a profit of exactly ₹ 2,00,000?" Set up the model with formulas, then Data → What-If Analysis → **Goal Seek**: Set cell = profit cell, To value = 200000, By changing cell = price.

Goal Seek model: which price makes profit = 2,00,000?

	A	B
1	**Input / output**	**Value**
2	Units sold	5000
3	Price	90.40
4	Variable cost per unit	42
5	Fixed costs	42000
6	Profit =B2*(B3-B4)-B5	200,000

```
B3    =50+40.4
B6    =B2*(B3-B4)-B5
```

Goal Seek finds the price ₹90.40: 5,000 × (90.40 − 42) − 42,000 = 2,00,000.

Data Tables: many scenarios at once

A one-variable or two-variable **Data Table** recalculates a model for a list of inputs. Example: profit for prices 80–100 (rows) and units 4,000–6,000 (columns).

STEP BY STEP **Two-variable data table**

1. Put the profit formula in the top-left corner cell of the grid (for example E2: =B6).
2. Prices down the column below it (E3:E7), units across the row to its right (F2:H2).
3. Select the whole grid (E2:H7) → What-If Analysis → **Data Table** → Row input cell = B2 (units), Column input cell = B3 (price).

Python

```python
prices = [80, 85, 90, 95, 100]; units = [4000, 5000, 6000]
grid = pd.DataFrame({u: [u * (p - 42) - 42000 for p in prices] for u in units},
index=prices)
grid.index.name = "Price ↓ / Units →"
grid
```

OUTPUT

	4000	5000	6000
Price ↓ / Units →			
80	110000	148000	186000
85	130000	173000	216000
90	150000	198000	246000
95	170000	223000	276000
100	190000	248000	306000

5 rows × 3 columns

Scenario Manager

Save named sets of inputs (Best case, Base case, Worst case) and switch between them or create a **Scenario Summary** report: What-If Analysis → Scenario Manager → Add.

Solver: optimisation

Solver (another free add-in) finds the best value of a target cell by changing several inputs, subject to constraints: for example, the product mix that maximises profit given limited machine hours and a minimum quantity for each product. Choose Simplex LP for linear problems.

LAMBDA: your own functions

`LAMBDA` (Microsoft 365) turns a formula into a reusable, named function. Define it once in Name Manager:

- Name: `GSTINCL`
- Refers to: `=LAMBDA(amount, rate, amount * (1 + rate))`

Then use it like any built-in function: `=GSTINCL(B2, 18%)`. Helper functions such as `MAP`, `REDUCE`, `SCAN`, `BYROW` and `BYCOL` apply a LAMBDA to every row or element of an array.

Python in Excel

Type `=PY(` in a cell (Microsoft 365, Formulas → Insert Python) and write pandas code. Python runs securely in Microsoft's cloud with Anaconda libraries (pandas, matplotlib, seaborn, scikit-learn, statsmodels):

```python
df = xl("tblSales[#All]", headers=True)
df.groupby("region")["sales"].sum().sort_values(ascending=False)
```

The result can be returned as an Excel value or a Python object, and charts appear as images in the cell. It is a great bridge to the Python track.

Macros, VBA and Office Scripts

Tool	Where it runs	Use for
Macro recorder	View → Macros → Record Macro	Recording repetitive formatting steps; saves as VBA
VBA	Desktop Excel (`Alt + F11`)	Custom automation, user forms; file must be `.xlsm`; macros can carry security risk, so enable only from trusted sources
Office Scripts	Excel for the web (Automate tab), TypeScript	Cloud automation; can be scheduled with **Power Automate**
Power Automate	Cloud	Refresh, email or save reports on a schedule without opening Excel

2026 UPDATE **Excel AI in 2026**

Copilot in Excel can generate formulas, highlight patterns, create pivot tables and charts, and run Python analysis from a plain-English prompt. Agentic "office agents" can plan multi-step tasks such as cleaning a file and building a summary. Use them to move faster, and always validate the output (AI for Data Analysis track, Lesson 4).

Summary

KEY TAKEAWAYS

- **Power Query** records cleaning steps and repeats them on Refresh: change types, split, replace, fill down, conditional columns, group by, **unpivot**, **merge** (join) and **append** (union).
- Its language is **M**; read it in the Advanced Editor.
- **Power Pivot / Data Model** stores related tables in a fast columnar engine; relationships replace VLOOKUP; **DAX measures** (SUM, DIVIDE, DISTINCTCOUNT, CALCULATE) compute correctly at every level.
- **Analysis ToolPak**: descriptive statistics, histogram, correlation, regression and tests (static output).
- **What-if**: Goal Seek (one input, target output), Data Tables (grid of scenarios), Scenario Manager (named sets), Solver (optimisation with constraints).
- **LAMBDA** creates custom functions; **Python in Excel** brings pandas into cells; macros/VBA and Office Scripts automate.

Quick quiz

6 questions · tap an answer to check it

1. Which Power Query command turns columns Jan, Feb, Mar into rows with Month and Sales columns?

A Pivot Column

B Unpivot Other Columns (with Region selected)

C Transpose

D Group By

Answer: B. Unpivot converts wide columns into attribute/value rows; "Other Columns" keeps working when new months are added.

2. Merge Queries in Power Query is equivalent to:

A A SQL JOIN or an XLOOKUP for every row

B A SQL UNION

C A pivot table

D Remove Duplicates

Answer: A. Merge joins two tables on matching key columns. Append is the UNION.

3. What is the main benefit of Power Query over manual cleaning?

A It is the only way to remove duplicates

B Steps are recorded and repeated automatically on Refresh

C It uses VBA

D It works only with CSV files

Answer: B. The recorded steps make the process repeatable and auditable.

4. Which tool finds the input value that produces a given result for one formula?

A Goal Seek

B Scenario Manager

C Data Table

D Descriptive Statistics

Answer: A. Goal Seek works backwards from a target output to one input.

5. Why is a DAX measure `DIVIDE([Total Profit],[Total Sales])` better than a pivot calculated field for margin?

A It is faster to type

B It is evaluated correctly in the current filter context at every level and handles division by zero

C Calculated fields cannot divide

D DAX works without a data model

Answer: B. Measures are context-aware and DIVIDE returns blank instead of an error for zero denominators.

6. The Analysis ToolPak reports mean 31,900 and median 8,400 for order values. What does this suggest?

A The data is symmetric

B The distribution is right-skewed by a few large orders

C There is an error in the data

D The mode is larger than the mean

Answer: B. When the mean is much larger than the median, a long right tail is pulling the mean up.

EXERCISE

Power Query practice

Using `video_game_sales.csv`: (1) load it with Power Query and set the correct types; (2) unpivot the four regional sales columns into `Region` and `Sales`; (3) group by Genre and Region to get total sales; (4) load the result to a pivot table and show % of row total.

Show solution (worked answer)

(1) Data → From Text/CSV → Transform Data; check that year is whole number and sales are decimal. (2) Select every column except na_sales, eu_sales, jp_sales and other_sales (or select those four and choose Unpivot Columns); rename Attribute → Region, Value → Sales; remove the global_sales column first so it is not unpivoted. (3) Home → Group By → Advanced: group by genre and Region, new column Sales = Sum of Sales. (4) Close & Load To → PivotTable Report; Rows genre, Columns Region, Values Sales → Show Values As % of Row Total. You should get the same mix as Lesson 8, Q3.

INTERVIEW CORNER

Questions you may be asked

Q: What is Power Query and why use it? Excel's (and Power BI's) ETL tool. It connects to many sources, applies recorded, repeatable transformations (types, splits, merges, unpivot, grouping) and refreshes with one click, which reduces manual errors.

Q: What is the difference between Merge and Append? Merge adds columns from another table by matching keys (a join); Append adds rows from tables with the same structure (a union).

Q: What is the Data Model and when do you need it? An in-memory columnar database inside Excel for related tables and large data. Use it when you have several tables, need distinct counts, or exceed normal sheet limits; it also supports DAX measures.

Q: How would you find the break-even price? Build the profit formula and use Goal Seek to set profit to 0 by changing price, or solve algebraically: price = variable cost + fixed costs ÷ units.

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